Genetics of common complex psychiatric disorders II

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Part 2: Molecular genetics

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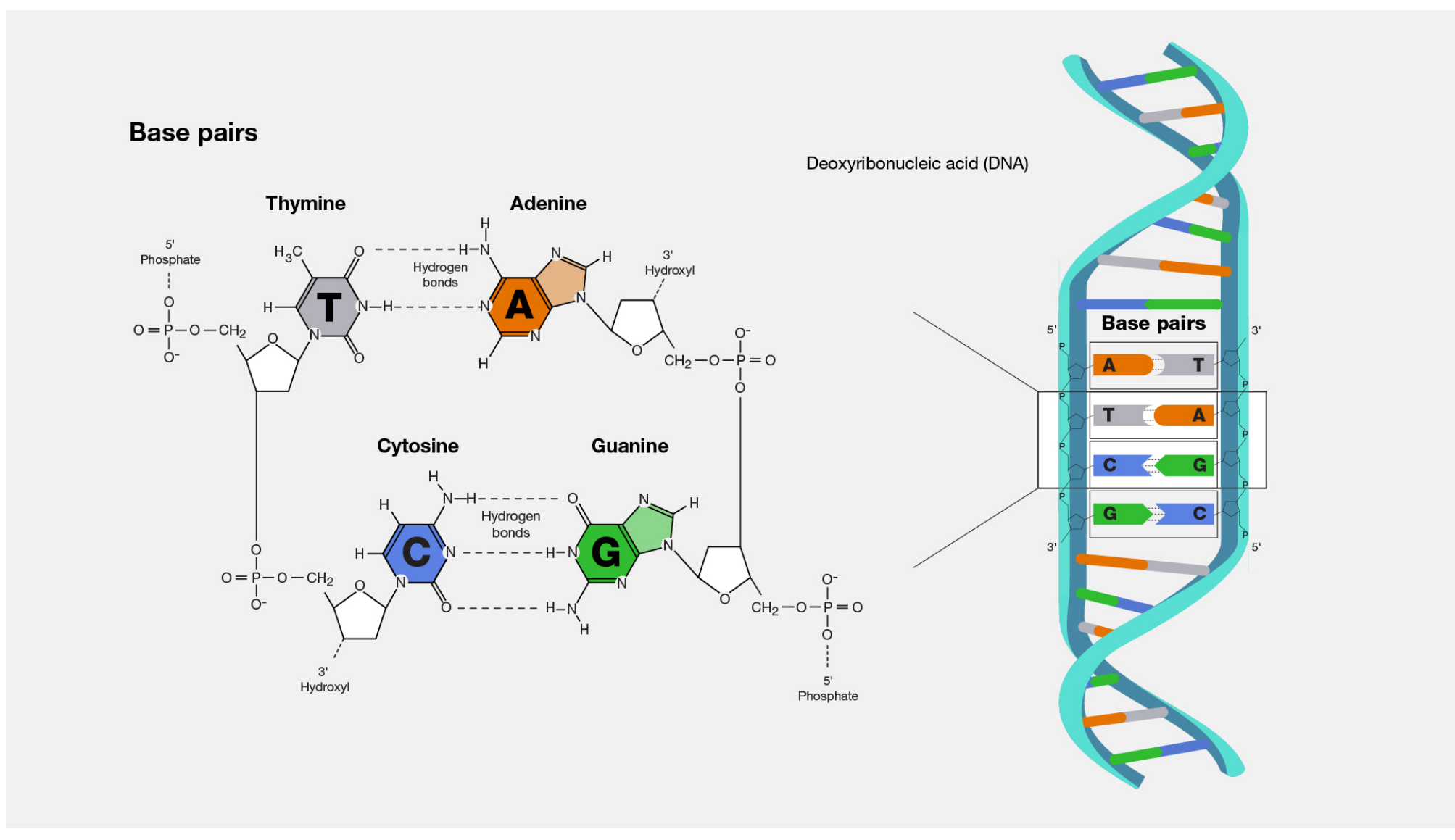
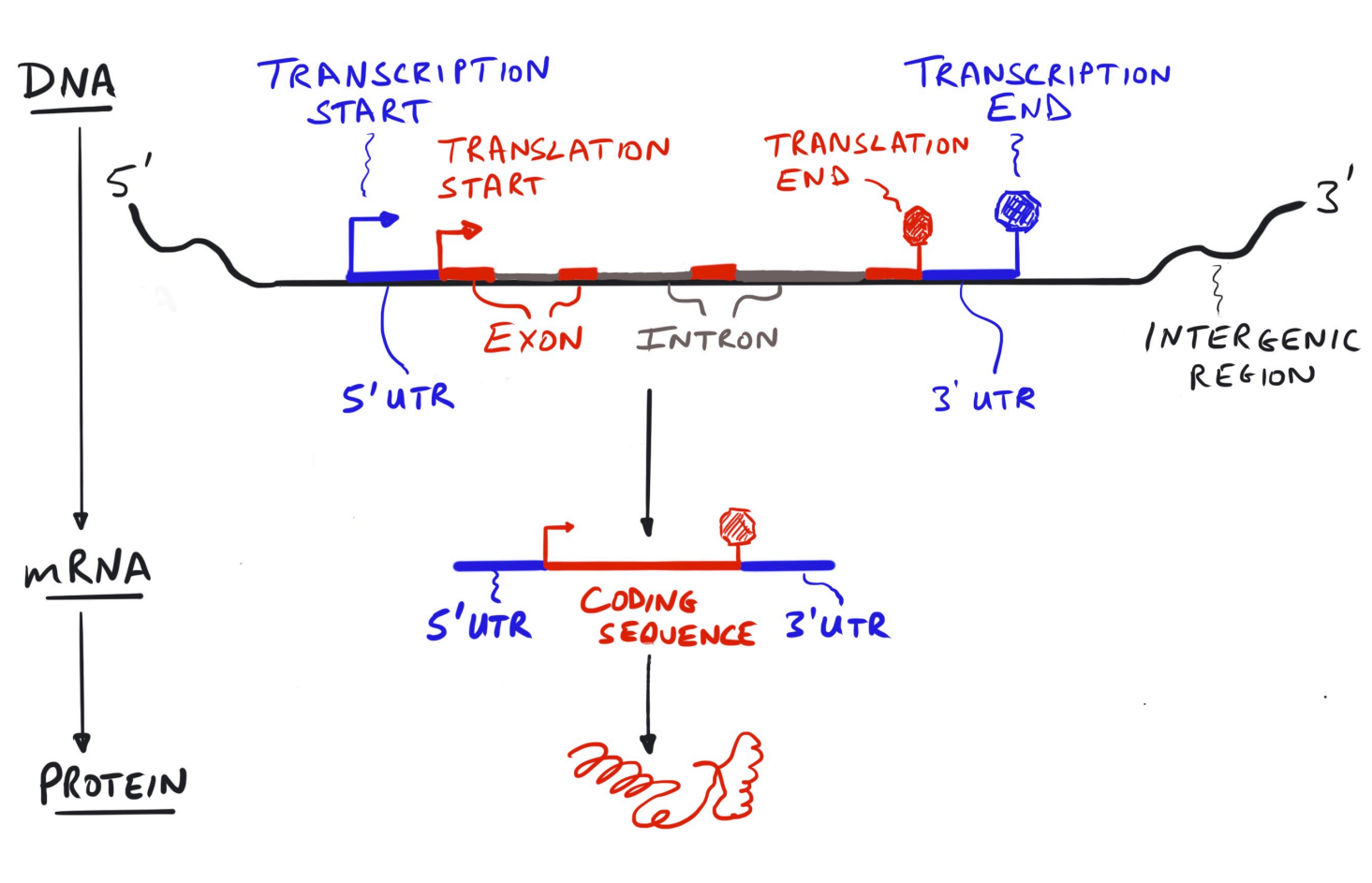
Genetics and Environmental Influences on Behaviour and Mental Health

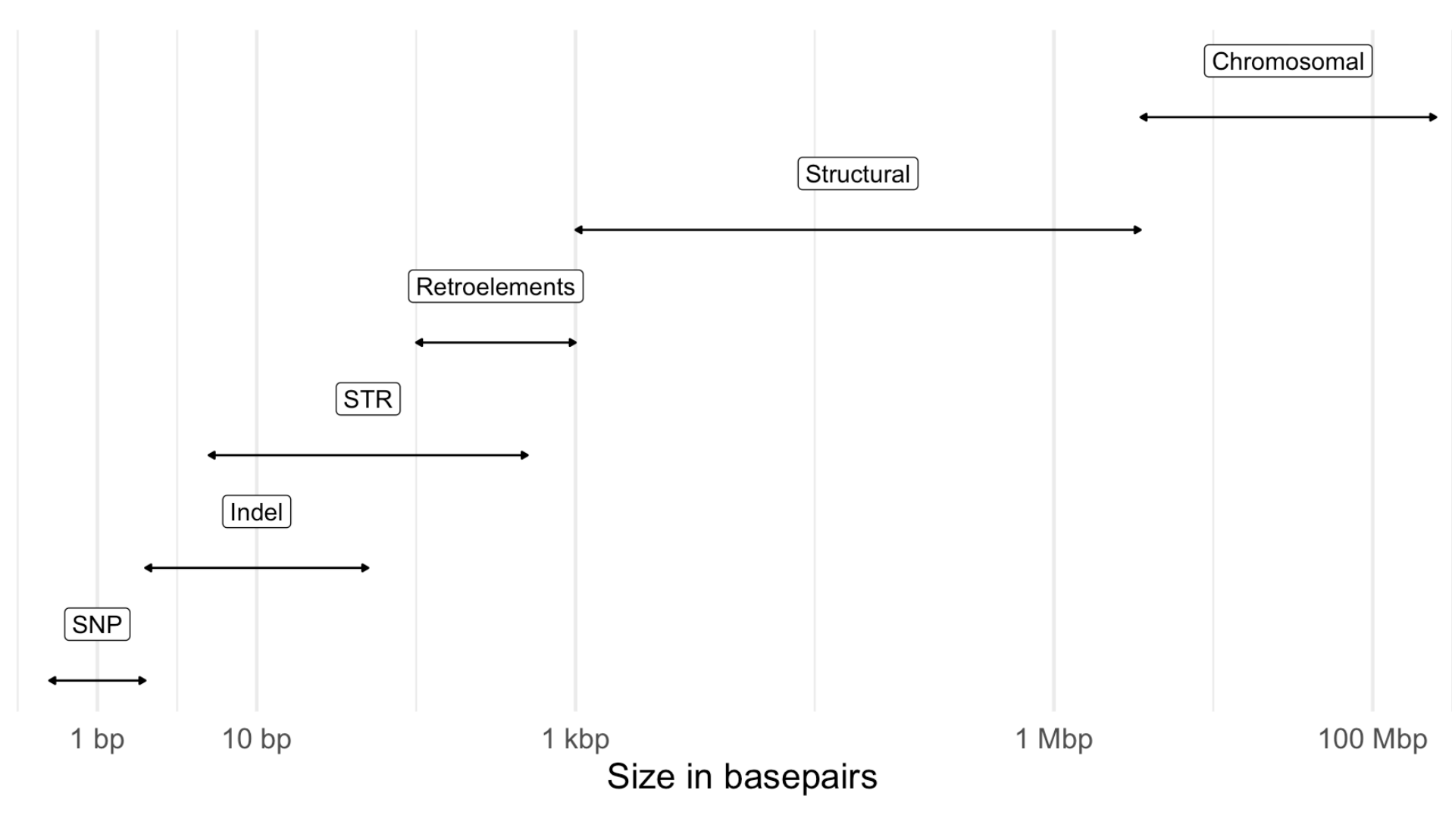
Structure of DNA


Base pairs
Thymine
Adenine
Cytosine
Guanine
Hydrogen bonds
5' Phosphate
3' Hydroxyl
T
A
C
G
Deoxyribonucleic acid (DNA)
Base pairs

Gene structure

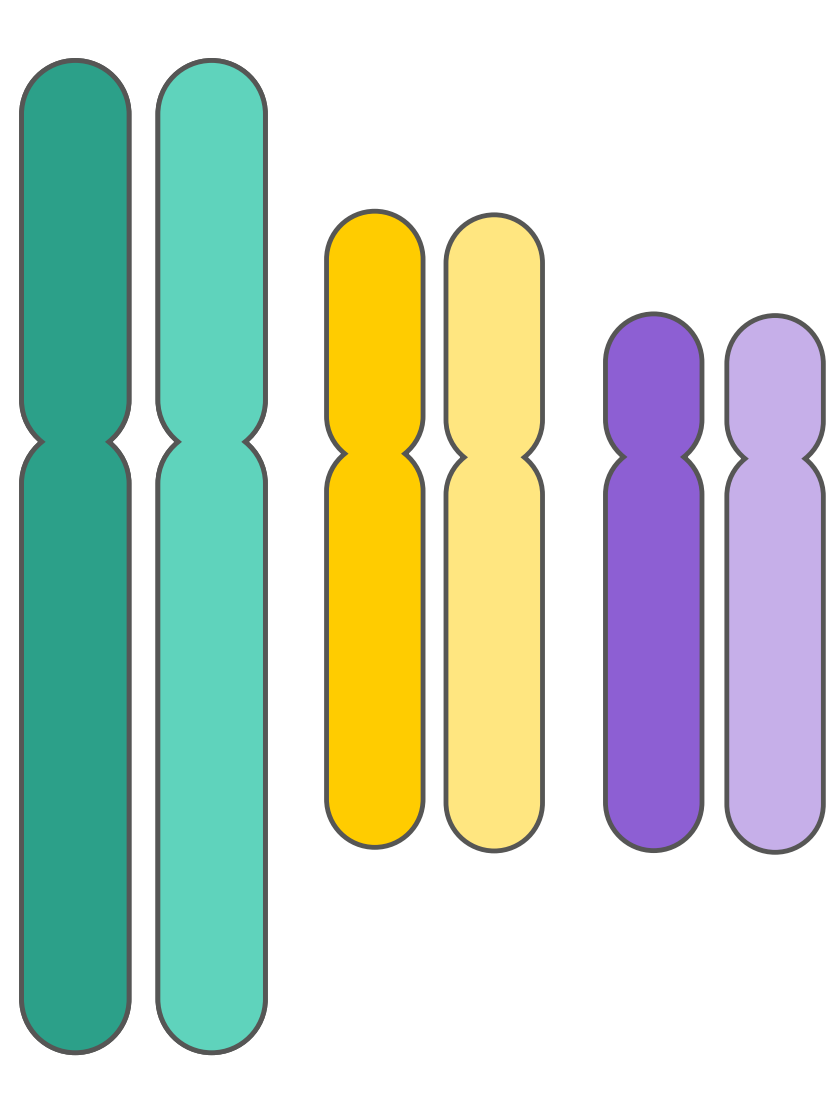

DNA
5'
3'
TRANSCRIPTION START
TRANSLATION START
TRANSLATION END
TRANSCRIPTION END
EXON
INTRON
5'UTR
3' UTR
INTERGENIC REGION
mRNA
5'UTR
CODING SEQUENCE
3'UTR
PROTEIN

Types and sizes of genetic variants

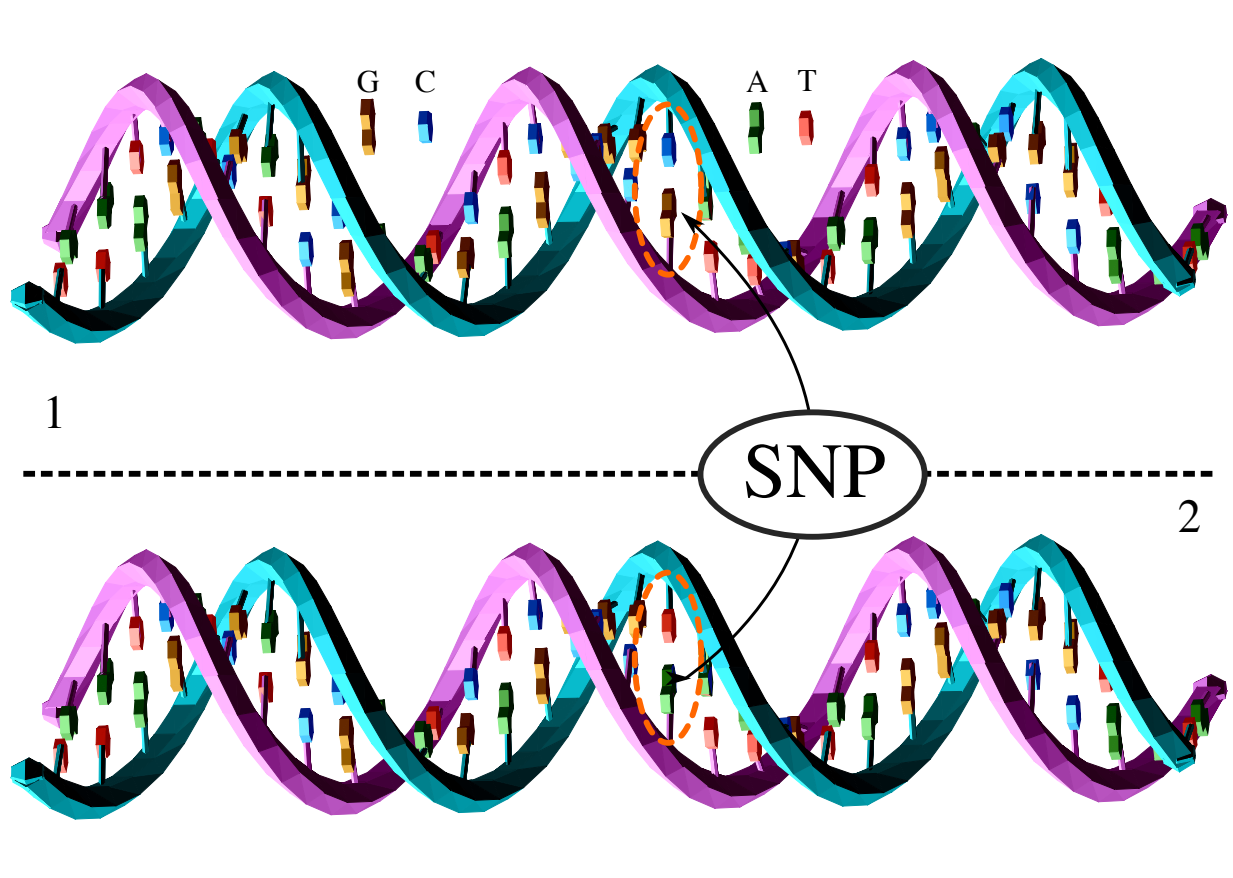


Chromosomes

Diploid (two copies of each chromosome)



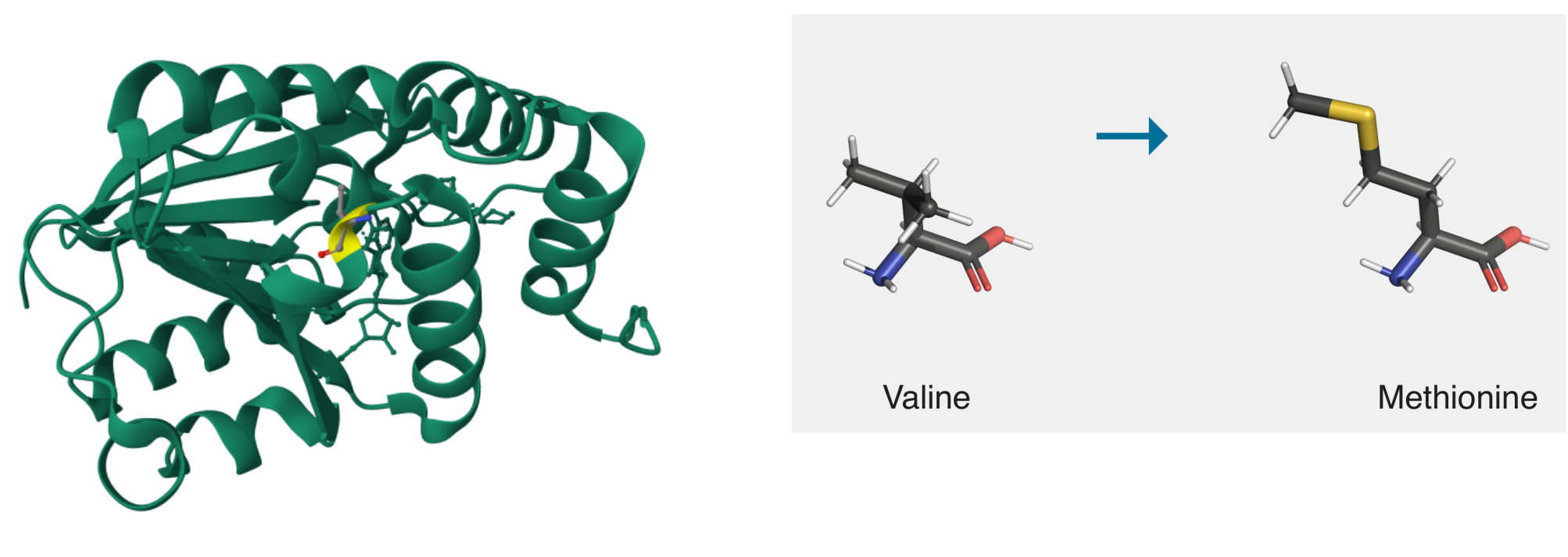
Single Nucleotide Polymorphisms (SNP)



- Single base of DNA is substituted with another.
- Occur in 1 of every 100 - 1000 basepairs (bp).
- Currently, ± 1.2B SNPs are known.

Candidate gene studies (circa 2000)

Example: Catechol-O-methyltransferase (COMT)


Valine
Methionine

SNP rs4680 G>A substitution (missense variant) results in amino acid change from valine to methionine at codon 158 (Val158Met).

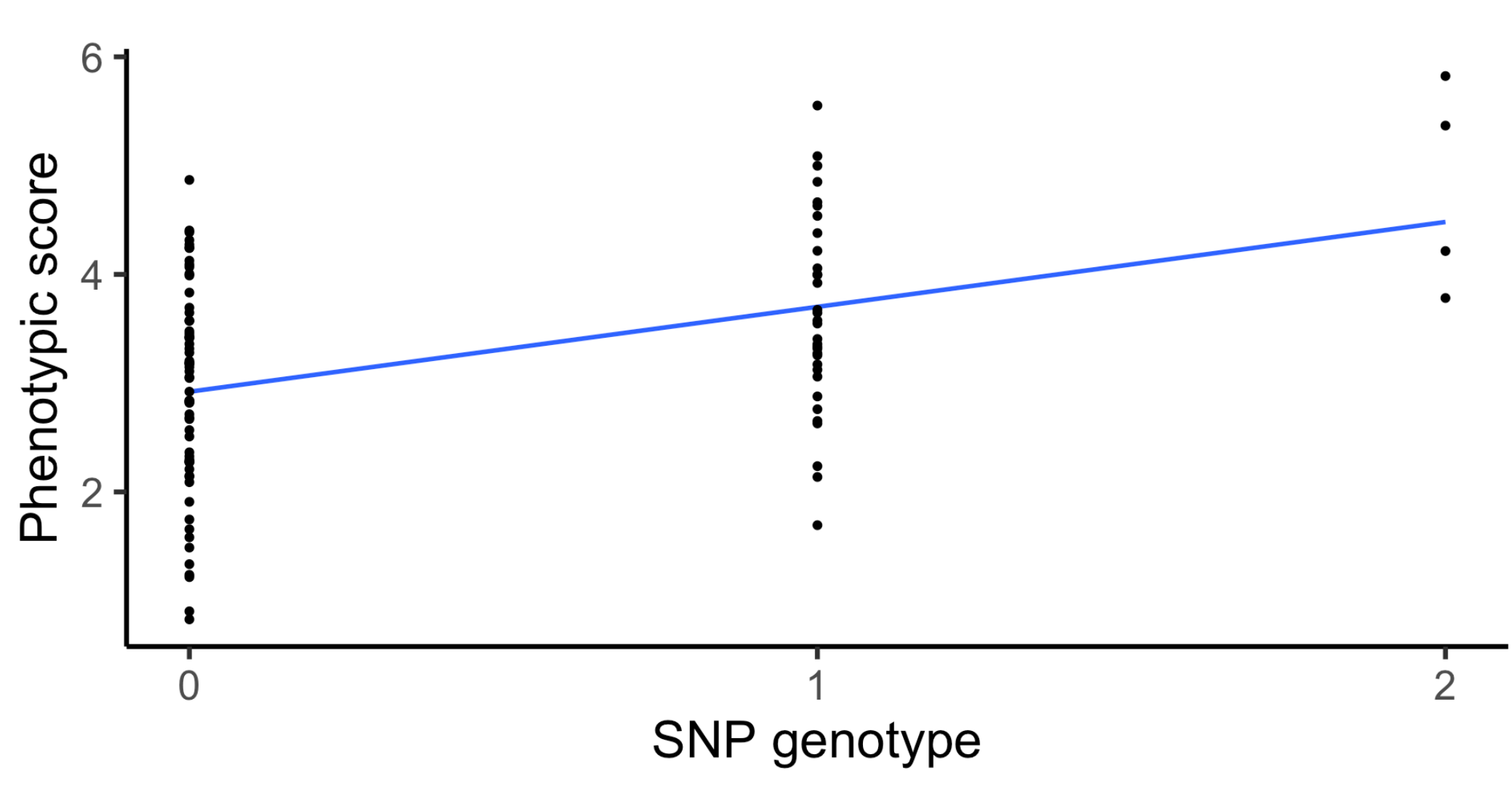
GWAS

Genome-wide associations studies (GWAS) (2007 - today)

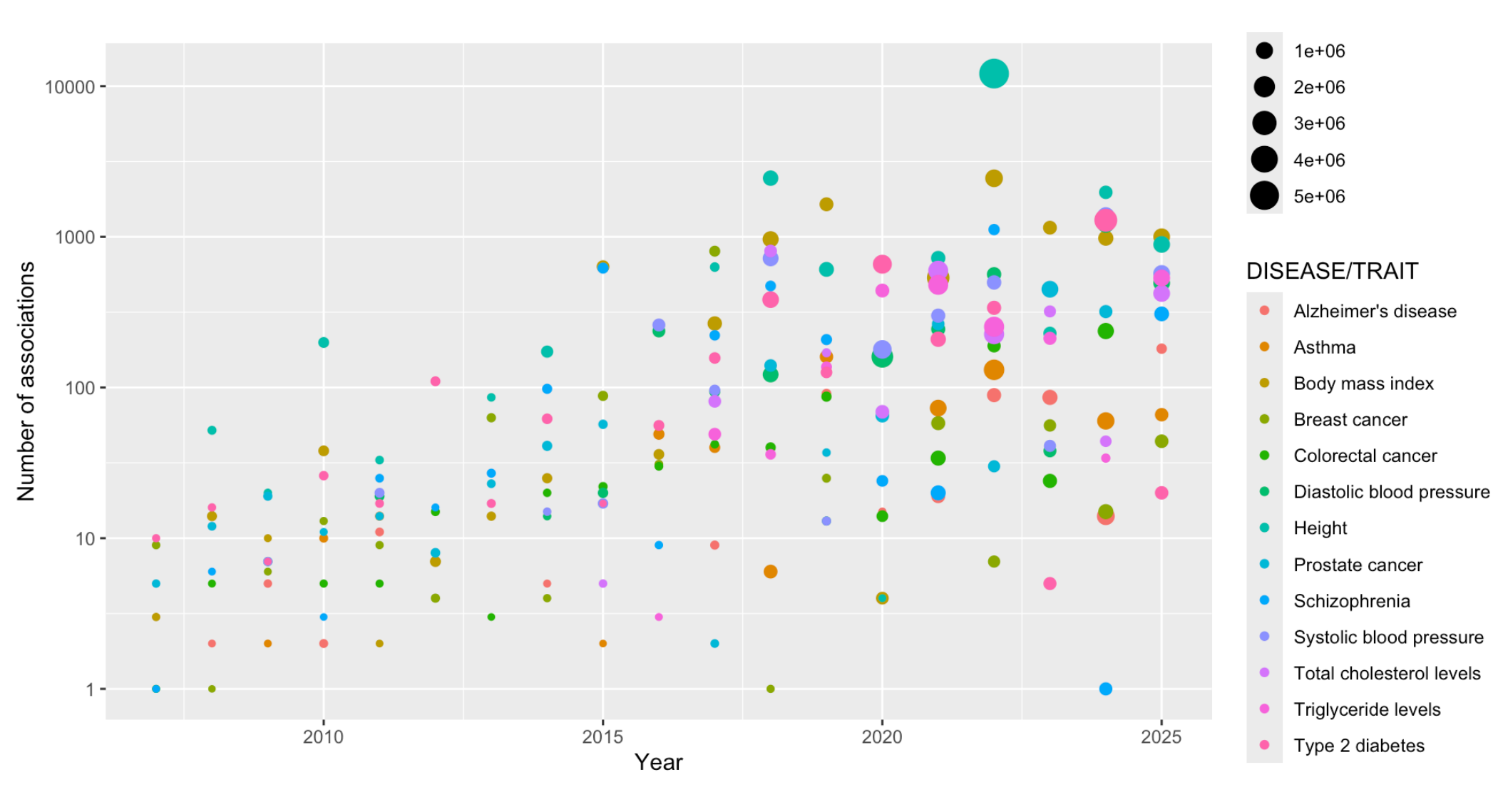
- Candidate gene findings did not replicate
- Power analysis: candidate gene study sample sizes were *vastly* too small
- Alternative: Assay the whole genome (hundreds of thousands or millions of variants)
- Hypothesis-free testing

Conducting a GWAS

- Two alleles: A and a
- Three genotypes: AA, Aa, aa
- Arbitrarily select one allele (A) as the *effect* allele
- For each subject, count number of effect alleles: $x = 0, 1, 2$



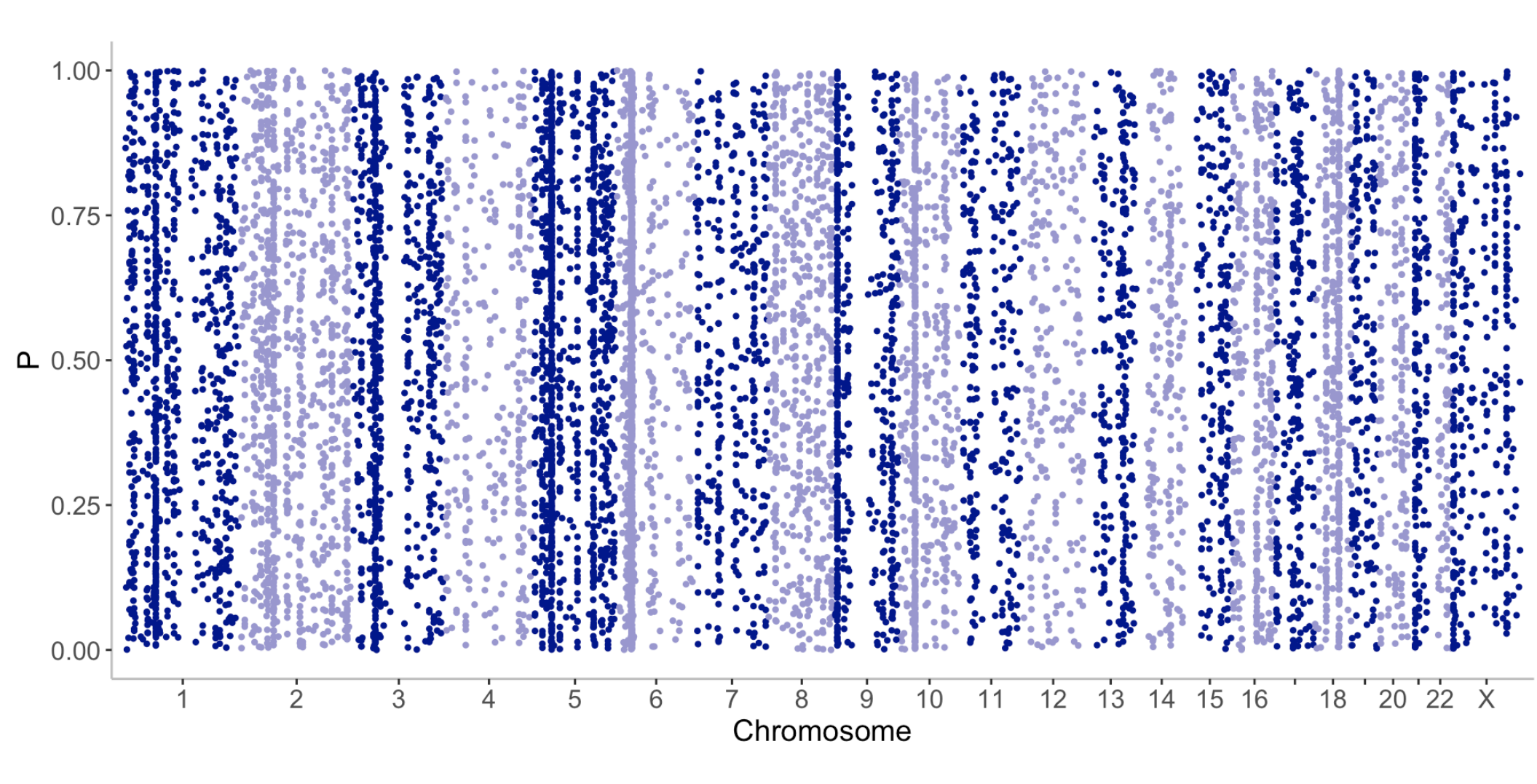
15+ years of GWAS



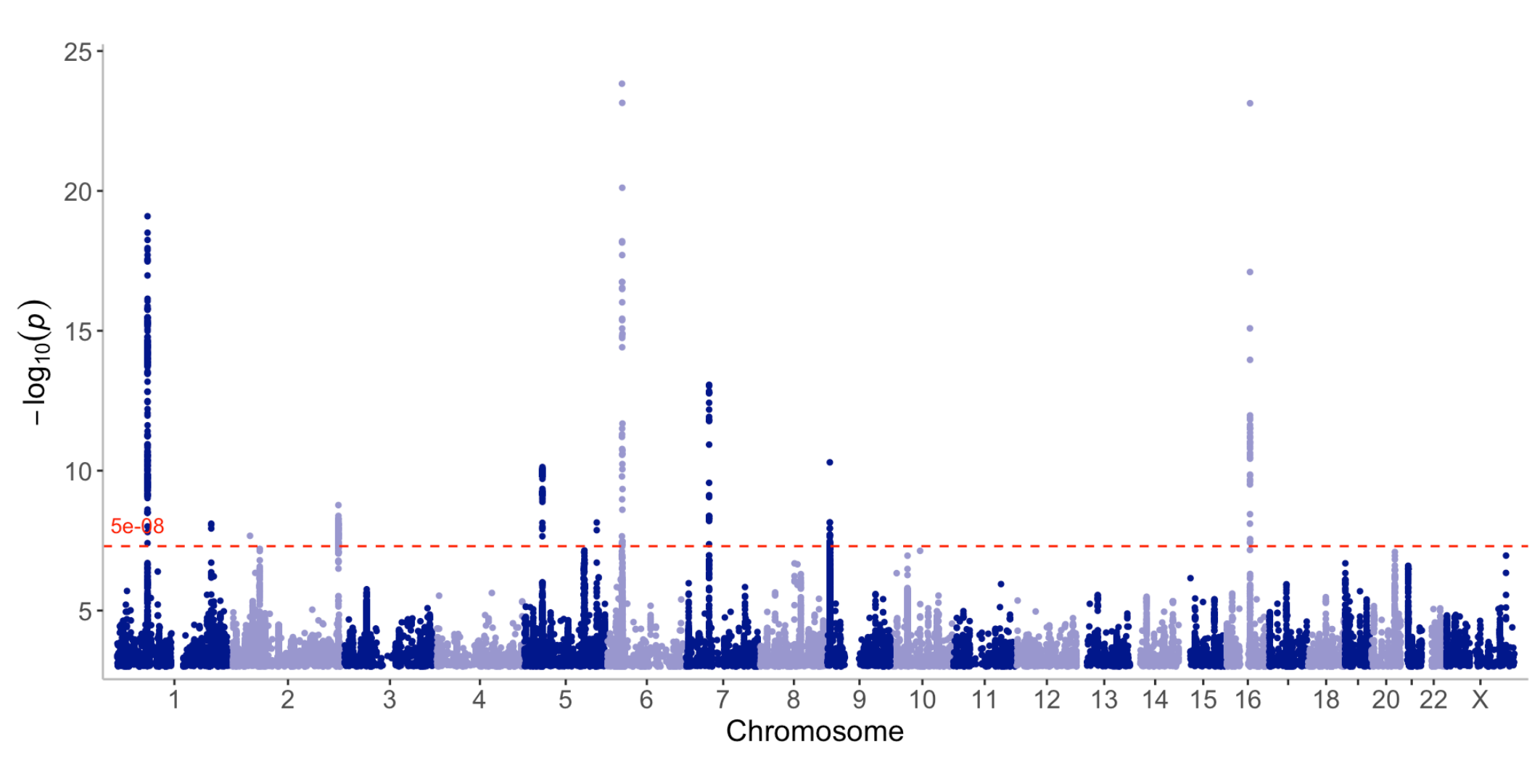
GWAS data

CHROM	POS	ID	REF	ALT	P	OR	AF
chr1	1006415	rs145588482	TGGCAGCTC	T	0.0004688	0.583384	0.01293
chr1	1007256	rs76233940	G	A	0.0004016	0.579783	0.01301
chr1	1341559	rs376494450	C	T	0.0001512	1.320130	0.02706
chr1	1480224	rs71628972	A	G	0.0007602	1.412750	0.01319
chr1	2035619	rs111334586	C	T	0.0009206	1.413370	0.01209
chr1	2213349	rs262695	A	G	0.0008815	0.904873	0.32073

Plotting GWAS results



Plotting GWAS results



Region plots

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[1] "Zoomed to region:  1:67038907-67359979"
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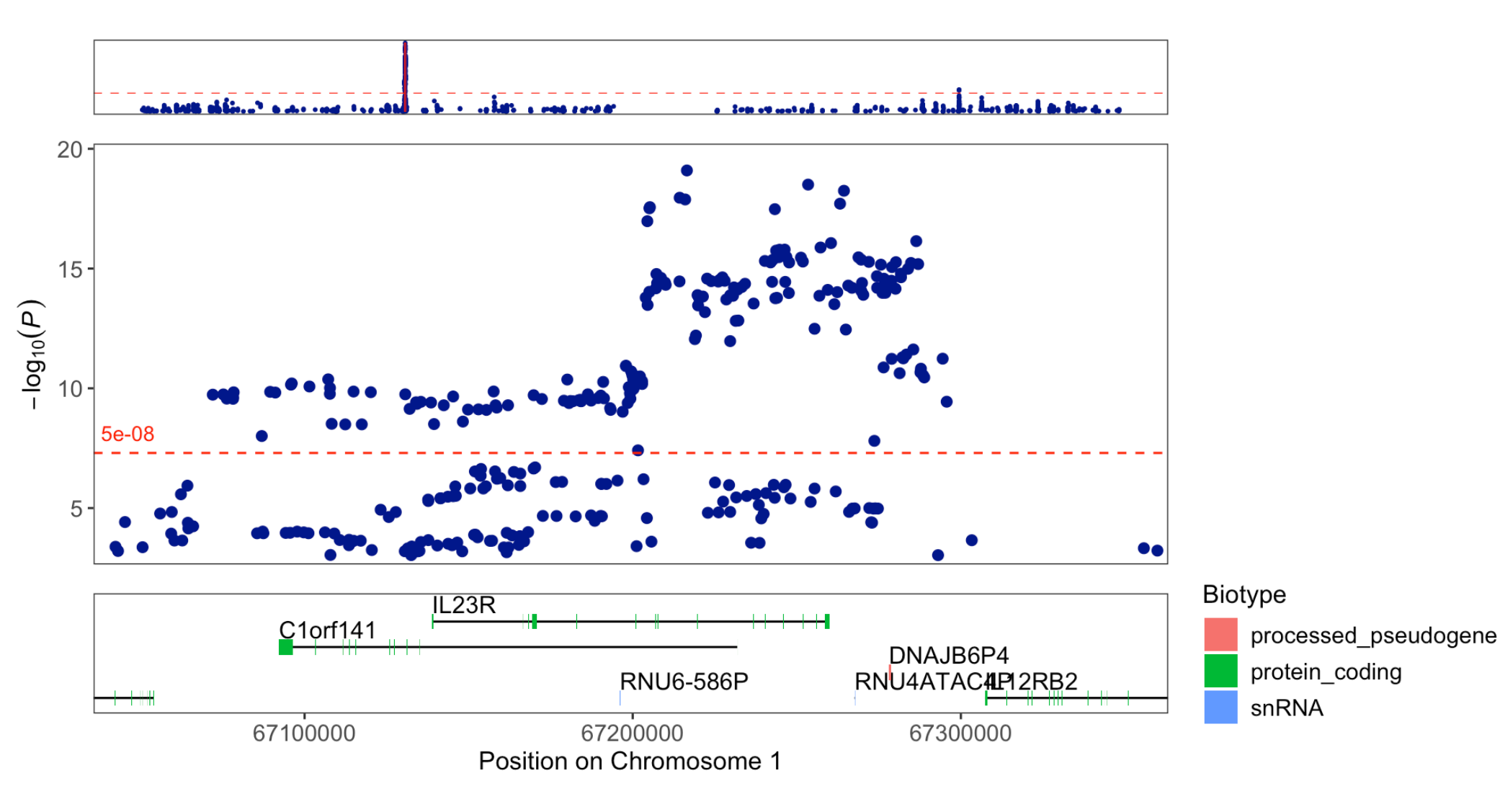

Linkage disequilibrium

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[1] "Zoomed to region:  1:67042284-67359988"
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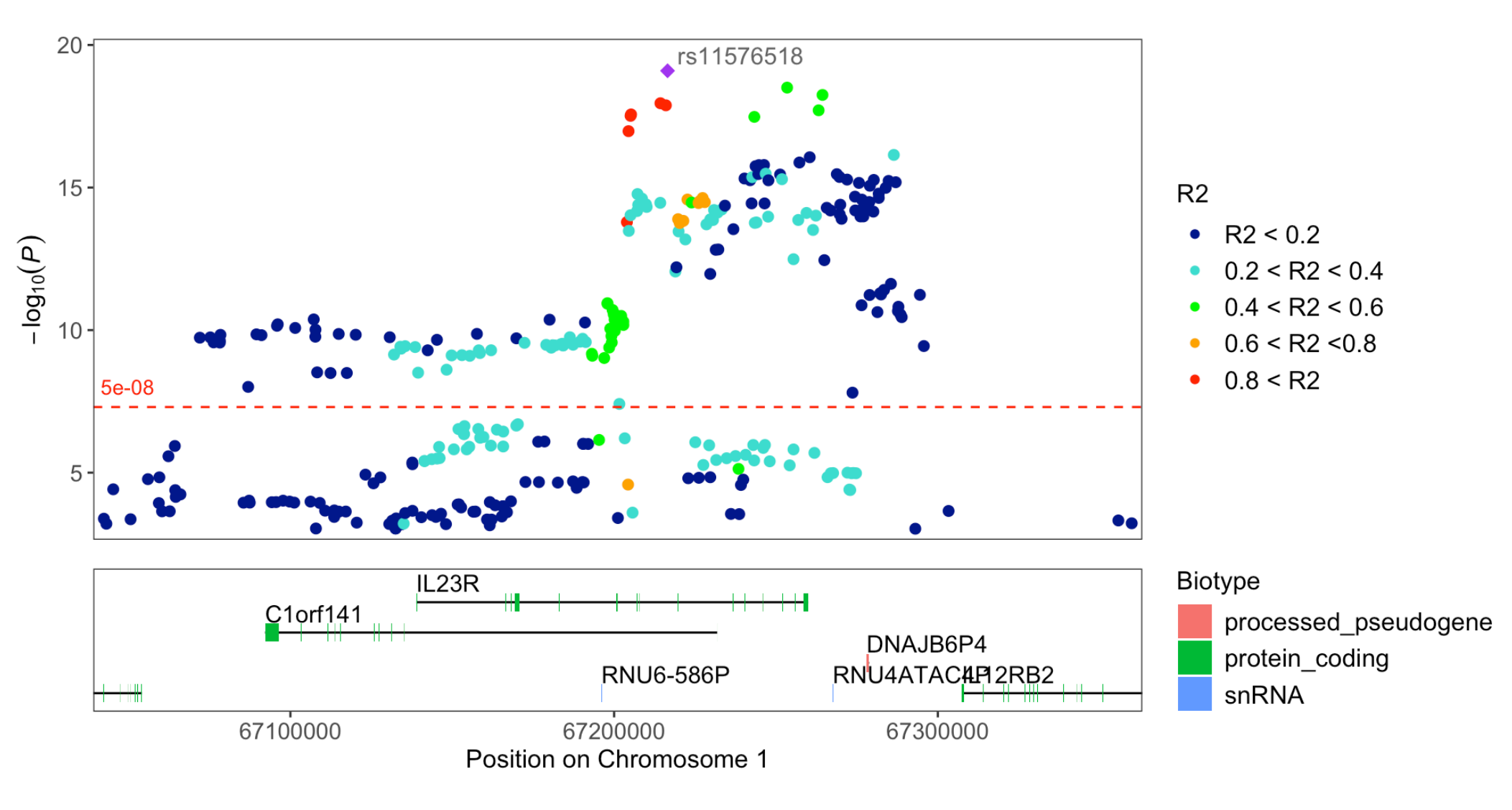


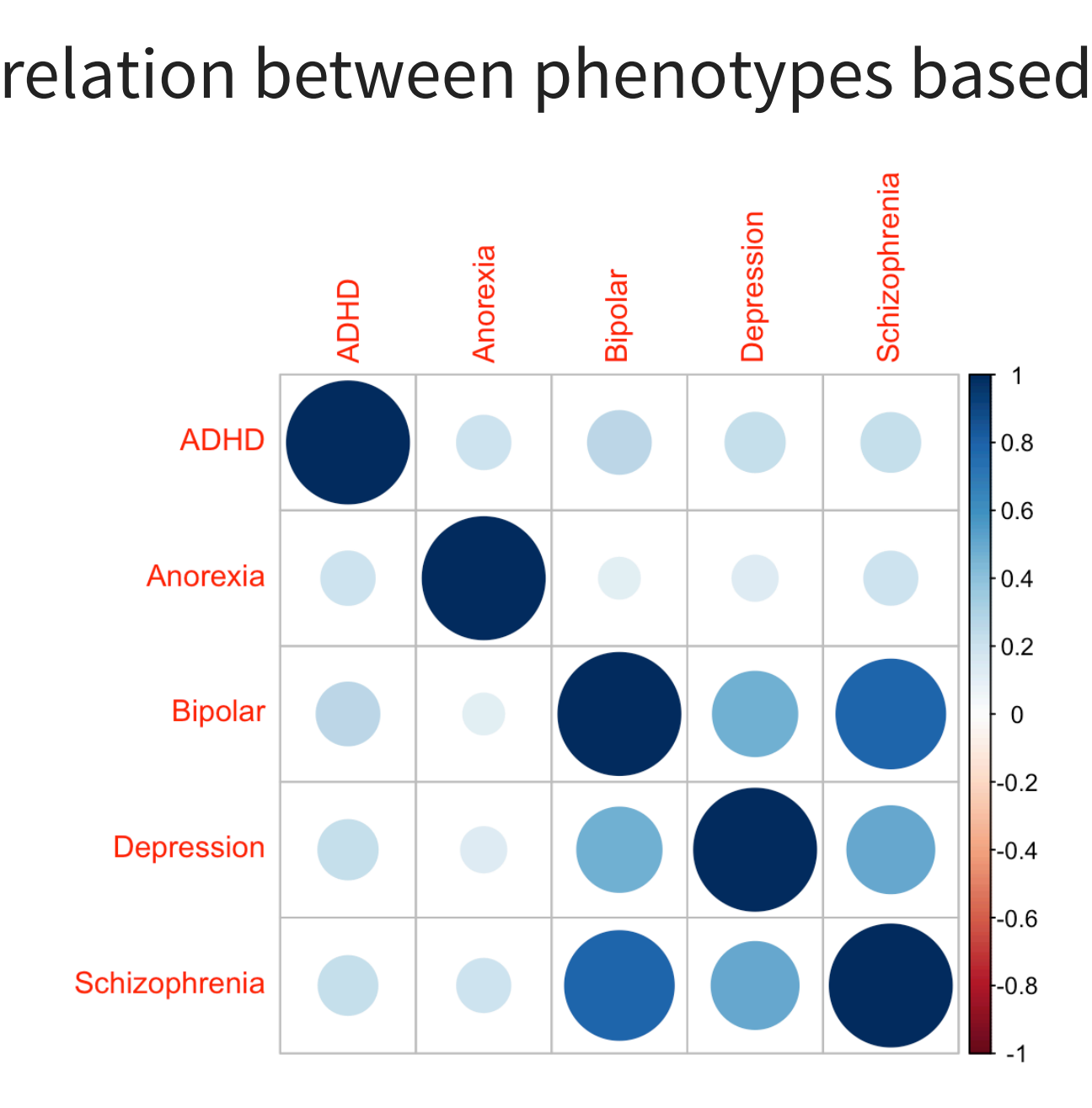
Key findings from psychiatric GWAS

Disorder	Key findings
Alcohol dependence	Genetically distinct from alcohol consumption but alcohol metabolism is key
Attention deficit/hyperactivity disorder	Common variants do not explain sex bias
Bipolar disorder	Calcium channel genes; neuronal hyperexcitability; Bipolar I ⇌ Schizophrenia; Bipolar II ⇌ Major depressive disorder
Major depression	Synaptic structure, neurotransmission; Prefrontal brain regions; Both ancestry and geography matter

Post-traumatic stress disorder	Higher heritability in women
Schizophrenia	Glutamatergic neurotransmission; Immunity; Disruptive coding variants

Crossing diagnostic boundaries

Genetic correlation between phenotypes based on GWAS data



Prediction

GWAS is a model

$$Y \sim a + b_i \times \text{SNP}_i + e$$

What if we reverse that?

$$\hat{Y} = \hat{a} + \hat{b} \times \text{SNP}$$

Polygenic score

$$\hat{Y} = \sum_i^n \hat{b}_i \times \text{SNP}_i$$

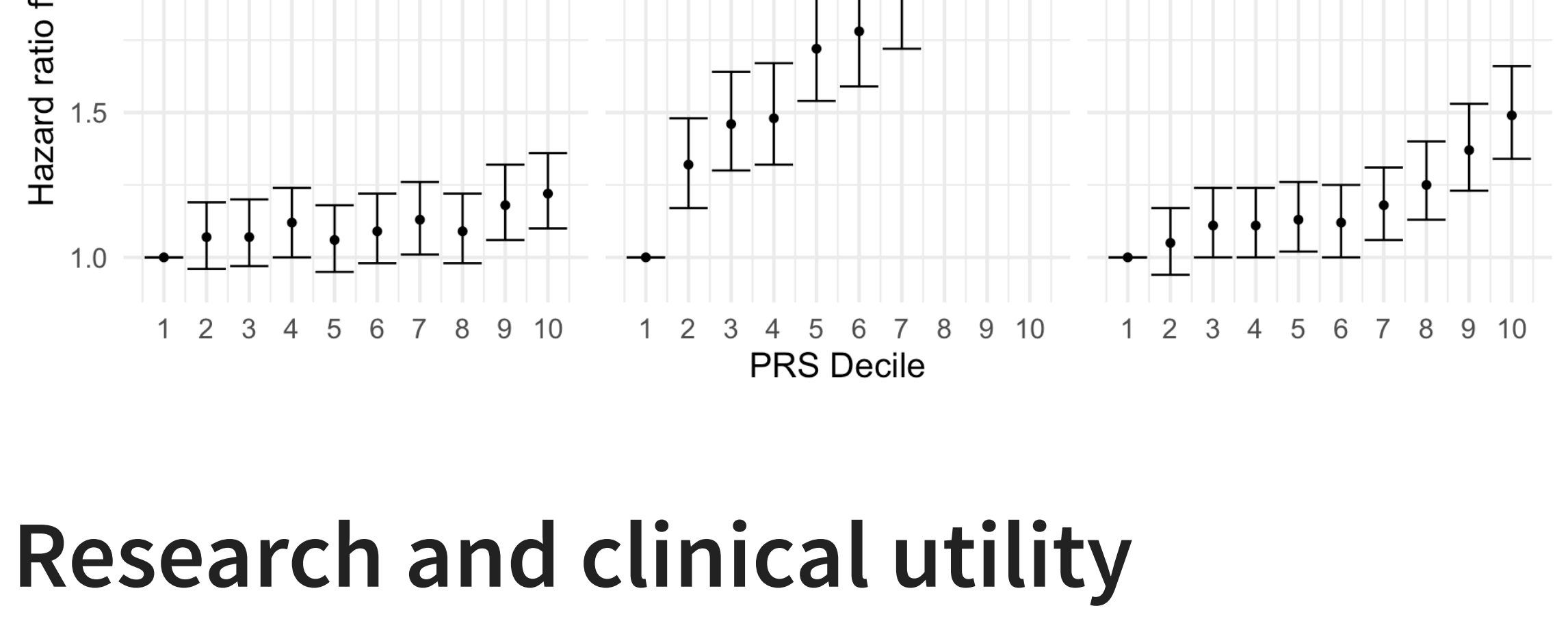
Calculating polygenic scores

SNP	Effect allele	Non-effect allele	Beta	P
rs61902807	C	T	-0.0352	1.45×10^{-44}
rs12142986	T	C	0.0318	3.38×10^{-41}
rs4702	G	A	0.0276	1.15×10^{-30}

At SNP rs61902807, the genotype contributions to the score would be

- TT: 0
- CT or TC: -0.0352
- CC: -0.0704

Polygenic prediction of major depression



Research and clinical utility

> "The accuracy with which a polygenic score can predict psychiatric illnesses, such as schizophrenia, bipolar disorder and major depression, is currently not sufficient for clinical use."

International Society of Psychiatric Genetics, *Advisory on the Use of Polygenic Risk Scores to Screen Embryos for Adult Mental Conditions.*

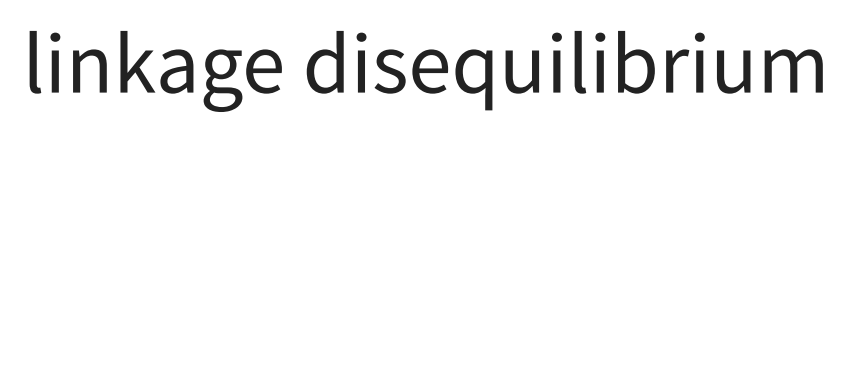
Causal analysis

Causality from genetics: Mendelian Randomisation

Mendel's Second Law of Independent Assortment is analogous to a randomised control trial, dubbed "Mendelian randomisation (MR).

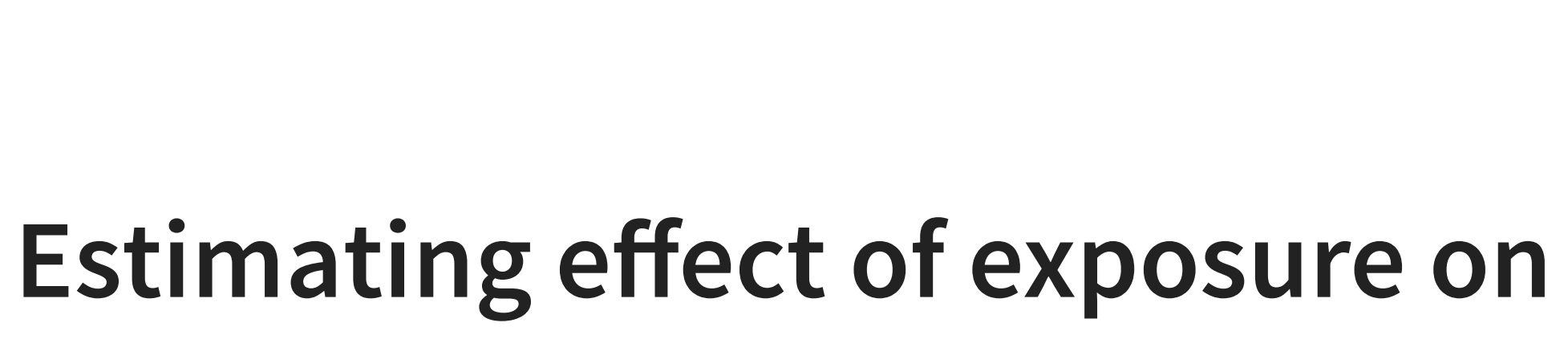
- Uses: Overcome confounding, reverse causation, selection bias, measurement attenuation
- Limitations: Reliable gene–phenotype link, pleiotropy, linkage disequilibrium

Genetic variant as an instrumental variable


Confounders

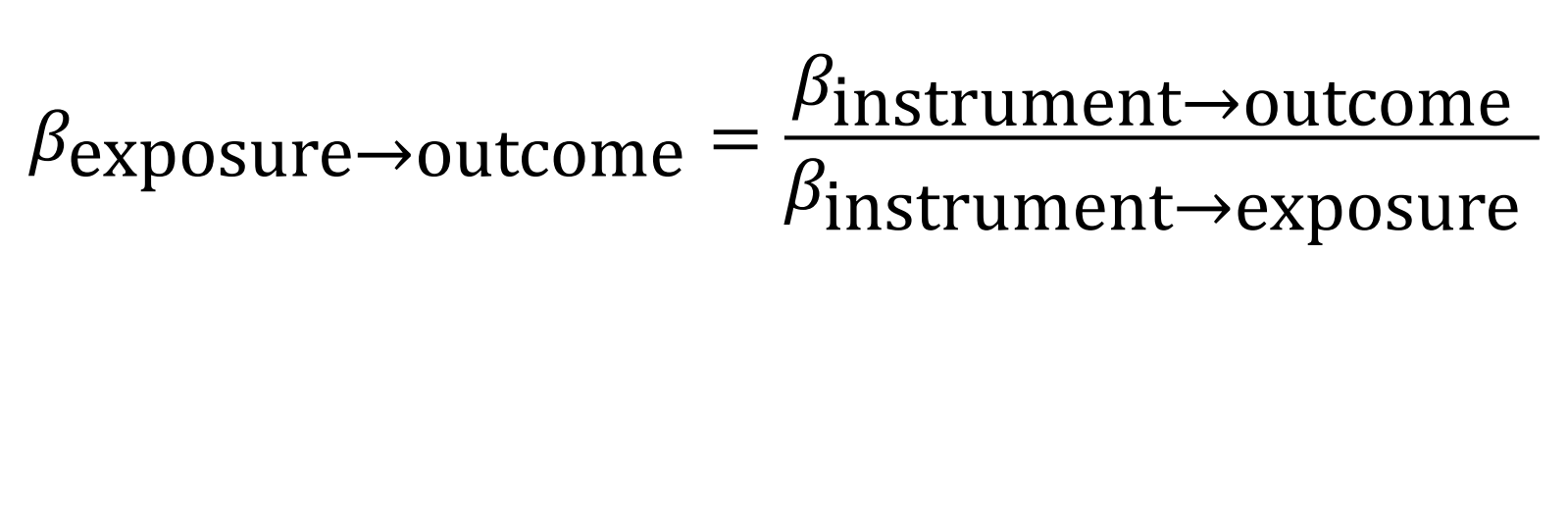
ADH1b → Exposure → Outcome

Genetic instrument from GWAS on exposure and outcome


Confounders

Genetic instrument → Exposure → Outcome

Estimating effect of exposure on outcome

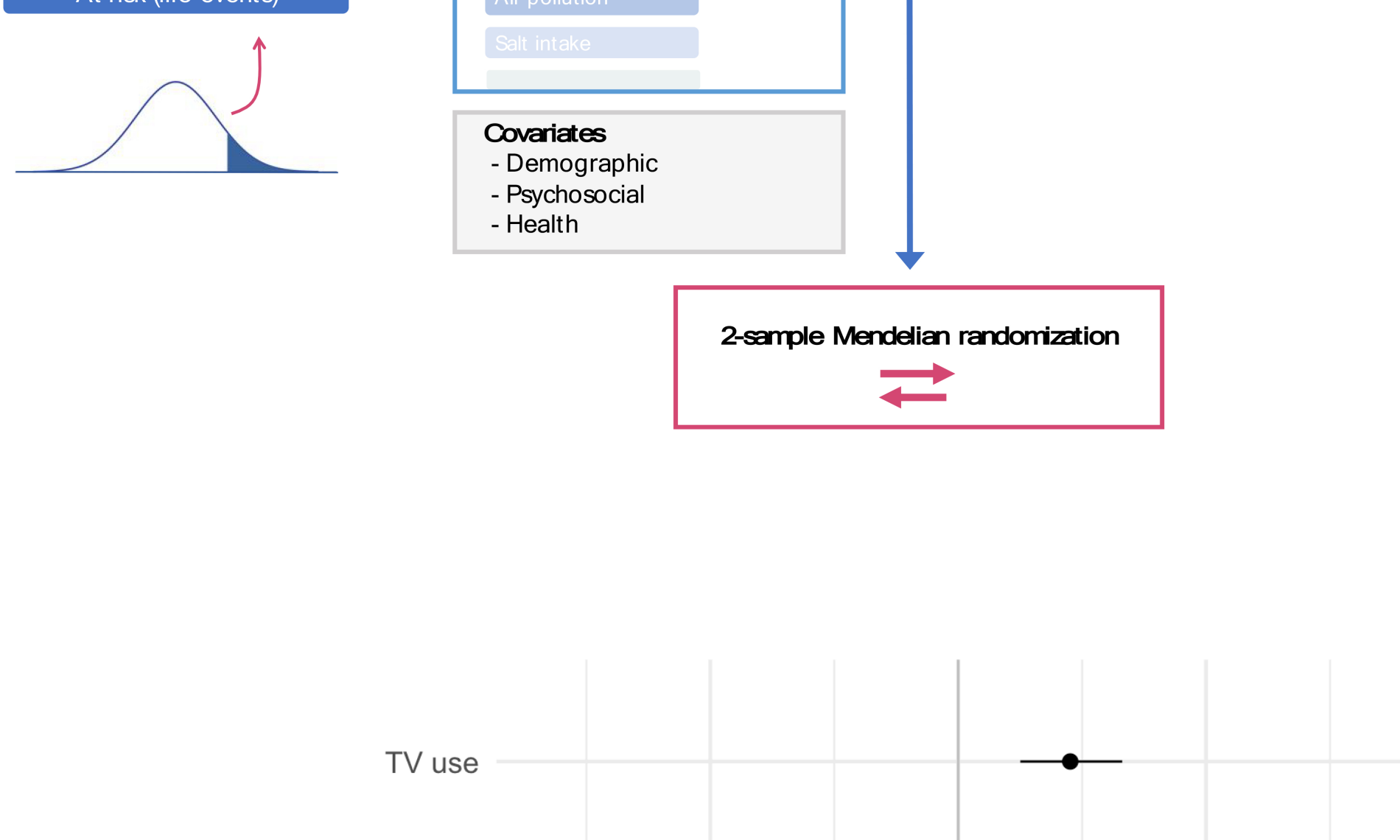

Confounders

Genetic instrument $\xrightarrow{\beta_{\text{instrument}\to\text{exposure}}}$ Exposure $\xrightarrow{\beta_{\text{exposure}\to\text{outcome}}}$ Outcome

$\beta_{\text{instrument}\to\text{outcome}}$

$$\beta_{\text{exposure}\to\text{outcome}} = \frac{\beta_{\text{instrument}\to\text{outcome}}}{\beta_{\text{instrument}\to\text{exposure}}}$$

Mendelian randomisation study: modifiable risk factors and depression


Baseline assessment/recruitment

Online survey follow-up

Analytic samples

- Full sample
- At risk (genetic)
- At risk (life events)

Modifiable factors

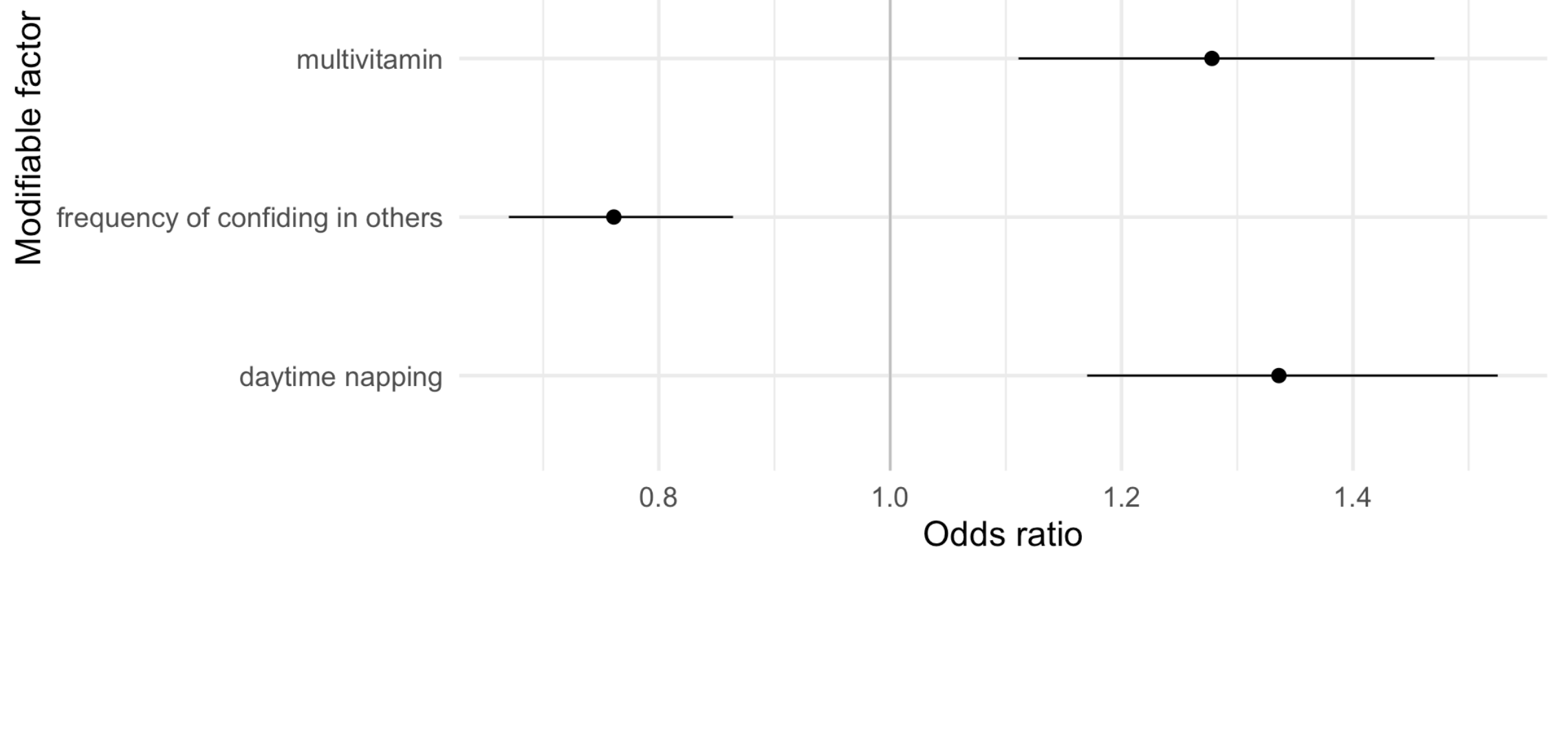
- TV time
- Emotional support
- Swimming or cycling
- Hours of sleep
- Air pollution
- Salt intake

Covariates

- Demographic
- Psychosocial
- Health

Clinically significant depression

2-sample Mendelian randomization



Resources